

Chunggi Lee

PHD CANDIDATE @ HARVARD UNIVERSITY

Science and Engineering Complex, 150 Western Ave, Boston, MA

☎ (+82) 10-4111-9393 | ✉ chungyi347@gmail.com & chunggi_lee@g.harvard.edu | 🌐 chungyi347.github.io

Education

Harvard University

PH.D IN COMPUTER SCIENCE

- GPA: 3.92/4.0, Advisor : [Prof. Hanspeter Pfister](#)

Boston, MA

Sep. 2023 - Now

Ulsan National Institute of Science and Technology

M.S IN ELECTRICAL AND COMPUTER ENGINEERING

- GPA : 4.17/4.3, Advisor : [Prof. Sungahn Ko](#)
- Thesis Title: GUIComp: A GUI Design Assistant with RealTime, MultiFaceted Feedback

Ulsan, South Korea

Mar. 2018 - Feb. 2020

Ulsan National Institute of Science and Technology

B.S IN ELECTRICAL AND COMPUTER ENGINEERING

- Magna Cum Laude
- Thesis Title: A Visual Analytics System for Exploring, Monitoring, and Forecasting Road Traffic Congestion.

Ulsan, South Korea

Mar. 2014 - Feb. 2018

Publications

INTERNATIONAL

- P1. **Chunggi Lee***, Hayato Saiki*, Tica Lin, Eiji Ikeda, Kenji Suzuki, Chen Zhu-Tian, Hanspeter Pfister, “**ViSTAR: Virtual Skill Training with Augmented Reality with 3D Avatars and LLM coaching agent**”, ACM CHI Conference on Human Factors in Computing Systems
- P2. Hayato Saiki, **Chunggi Lee**, Hikari Takahashi, Tica Lin, Hidetada Kishi, Kaori Tachibana Ibaraki, Yasuhiro Suzuki, Hanspeter Pfister, Kenji Suzuki, “**BRIDGE: Borderless Reconfiguration for Inclusive and Diverse Gameplay Experience via Embodiment Transformation.**”, ACM CHI Conference on Human Factors in Computing Systems (**Best Paper Award**)
- P3. **Chunggi Lee**, Ut Gong, Tica Lin, Stefanie Zollmann, Scott A Epsley, Adam Petway, Hanspeter Pfister, “**VAIR: Visual Analytics for Injury Risk Exploration in Sports.**”, IEEE Transactions on Visualization and Computer Graphics (**IEEE VIS VAHC Workshop**), 2025.
- P4. **Chunggi Lee**, Tica Lin, Hanspeter Pfister, Zhu-Tian Chen, “**Sportify: Question Answering with Embedded Visualizations and Personified Narratives for Sports Video.**”, IEEE Transactions on Visualization and Computer Graphics (**IEEE VIS**), 2024. [\[PDF\]](#) [\[Webpage\]](#)
- P5. Namhyuk Ahn, Junsoo Lee, **Chunggi Lee**, Kunhee Kim, Daesik Kim, Seung-Hun Nam, Kibeom Hong, “**DreamStyler: Paint by Style Inversion with Text-to-Image Diffusion Models.**”, The Association for the Advancement of Artificial Intelligence (**AAAI**), 2024. [\[PDF\]](#) [\[Webpage\]](#) [\[Code\]](#)
- P6. Han Kim*, **Chunggi Lee***, Junsoo Lee*, Dohyun Kim, Kwangjin Lee, Moohyun Oh, Daesik Kim, “**FlatGAN: A Holistic Approach for Robust Flat-Coloring in High-Definition with Understanding Line Discontinuity.**”, ACM Multimedia (**MM**), 2023. [PDF](#)
- P7. Cholmin Kang, **Chunggi Lee**, Heon Song, Minuk Ma, Sergio Pereira, “**Variability Matters: Evaluating inter-rater variability in histopathology for robust cell detection.**”, European Conference on Computer Vision Workshop On AI-Enabled Medical Image Analysis (**ECCVW**), 2022. [\[PDF\]](#)
- P8. **Chunggi Lee**, Seonwook Park, Heon Song, Jeongun Ryu, Sanghoon Kim, Haejoon Kim, Sergio Pereira, Donggeun Yoo, “**Interactive Multi-Class Tiny-Object Detection.**”, IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2022. [\[PDF\]](#) [\[VIDEO\]](#)
- P9. Cheonbok Park, **Chunggi Lee**, Hyojin Bahng, Yunwon Tae, Kihwan Kim, Seungmin Jin, Sungahn Ko, Jaegul Choo, “**ST-GRAT: A novel spatio-temporal graph attention networks for accurately forecasting dynamically changing road speed.**”, ACM International Conference on Information and Knowledge Management (**CIKM**), 2020 (20.9% acceptance rate). [\[PDF\]](#)
- P10. **Chunggi Lee**, Sanghoon Kim, Dongyun Han, Hongjun Yang, Young-Woo Park, Bum Chul Kwon, Sungahn Ko, “**GUIComp: A GUI Design Assistant with RealTime, MultiFaceted Feedback**”, ACM CHI Conference on Human Factors in Computing Systems (**CHI**), 2020, Accepted. [\[PDF\]](#) [\[Preview Video\]](#) [\[VIDEO\]](#)
- P11. **Chunggi Lee**, Yeonjun Kim, Seungmin Jin, Dongmin Kim, Ross Maciejewski, David Ebert, and Sungahn Ko, “**A Visual Analytics System for Exploring, Monitoring, and Forecasting Road Traffic Congestion.**” IEEE transactions on visualization and computer graphics (**TVCG** IF=4.579), 2019 (Proc. **IEEE VIS’19**), Accepted. [\[LINK\]](#) [\[PDF\]](#) [\[VIDEO\]](#) [\[NEWS1\]](#) [\[NEWS2\]](#) [\[NEWS3\]](#)

PREPRINT, POSTER, DOMESTIC

- D1. Hyunwook Lee, **Chunggi Lee**, Hongkyu Lim, Sungahn Ko, "TILDE-Q: A Transformation Invariant Loss Function for Time-Series Forecasting", (Under Review), 2022.
- D2. Juyoung Oh, **Chunggi Lee**, Hwiyeon Kim, Kihwan Kim, Osang Kwon, Eric D. Ragan, Bum Chul Kwon, Sungahn Ko, "An Empirical Study on the Relationship Between the Number of Coordinated Views and Visual Analysis.", Arxiv. [PDF]
- D3. **Chunggi Lee**, Juyoung Oh, Seungmin Jin, Isaac Cho, and Sungahn Ko, "A Graphical Workflow Exploration Environment For Visual Analytics.", Arxiv. [PDF]
- D4. Kihwan Kim, Sanghoon Kim, **Chunggi Lee**, Sungahn Ko, "Modeling Exploration/Exploitation Decisions through Mobile Sensing for Understanding Mechanisms of Addiction.", (MobiSys), 2019 (Poster)
- D5. J. Lee, K. Kim, **C. Lee**, and S. Ko, "Visualization based Deep Learning Analysis Technology." The Korean Society for Noise and Vibration Engineering (KSNVE), 2017, Accepted. [LINK] [PDF]
- D6. Y. Oh, **C. Lee**, J. Oh, J. Yang, H. Gwag, S. Moon, S. Park, and S. Ko, "Introduction of Visual Analytics." Korea Computer Graphics Society, 2016, Accepted. [LINK] [PDF]

Professional Experience

Naver Webtoon AI.

Gyeonggi-do, South Korea

RESEARCHER SCIENTIST (ALTERNATIVE MILITARY SERVICE) [LINK]

Aug 2022 - Aug 2023

- Conducted research on **stable diffusion models** to support the process of webtoon (comics) creation
- Designed and built an **AI platform** frontend and backend to distribute deep learning models for users.

Lunit Inc.

Seoul, South Korea

RESEARCHER ENGINEER (ALTERNATIVE MILITARY SERVICE) [LINK]

Mar 2020 - Aug-2022

- Designed and built interactive deep learning models for reducing cost of annotations (e.g., **object detection** and **segmentation**). (P2)
- Implemented an annotation tool and an internal operation tool with **Django, DRF, React, and mui**.
- Deployed and served deep learning models by using **TorchServe and Fast API**.
- Analyzed quality of pathology data which have significant different among experts and proposed a new method for evaluating pathology large-scale dataset (P1)

Interactive Visual Analysis & Data Exploration Research Lab

Ulsan, South Korea

RESEARCHER & DEVELOPER [LINK]

Jun 2016 - Mar 2020

- Built interactive tools and visual analytic systems for **mobile GUI design authoring for novice** (P4), **traffic congestion forecasting and propagation** (P5), **multiple coordinate visualization** (D3), and **gene expression** (D4).
- Implemented **spatio-temporal deep learning models** (P3), mobile GUI recommendation deep learning, biomedical data clustering

Honors & Awards

2017 **4th Place & Bronze Prize**, National Super Computing Competition

Ulsan, South Korea

2017 **3rd Place & Bronze Prize**, Naver UNIST Undergraduate Poster Award '17 UNIST \$1,500 [POSTER]

& KISTI & UNIST

2016 & **Semester GPA 3.94/4.3, GPA 3.93/4.3**, Semester Academic Excellence Award for 2016 Fall Semester, 2017

Naver & UNIST

2017 Fall Semester

UNIST

Patent

- Graphic User Interface Assistant Device and Method in Mobile Environment** – Application Number : 10-2019-0001190
- Traffic Information Visualization Analysis Device and Method** – Application Number : 10-2018-0079727 (Acquired - \$0.1M)

Services

Reviewer, ACM CHI (2024, 26), IEEE VIS (2023-25), CVPR (2026), PacificVis (2026), TVCG(2026), DIS (2026), TGRS (2026)

2019, 24 **Student Volunteer**, IEEE VIS

Vancouver, Canada

2015, 19 **Teaching Assistant**, Object Oriented Programming

Ulsan, South Korea

Mentoring

2025 **Mentee**, Hayato Saiki (University of Tsukuba), **Nina Doerr** (University of Stuttgart), **Max Wagner (HCRP Award)** (Harvard University), **Alexandre Ben Ahmed Kontoul** (EPFL)

Boston, MA

2024 **Mentee**, **Arthur Andre - MS Thesis (6.0/6.0)** (EPFL -> Amazon), **Zhuo Wang** (Xi'an Jiaotong-Liverpool University), **Ut(Jo Jo) Gong** (University of Washington)

Boston, MA